

Human-Robot Friendship (HRF) Procedure with Children

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Abstract

Intergroup contact theory posits that a positive encounter with a member of another social group (i.e., outgroup) can improve the individual's attitudes toward the entire outgroup [3] and even non-contacted outgroups [34]. The current study applies the *human-robot friendship (HRF)* procedure [5] to children in early ($N = 68$) and middle childhood ($N = 60$), to investigate how a brief, positive contact with a social robot affects their perceptions of robots and other human outgroups. The HRF procedure indeed improved attitudes toward social robots in both age groups. However, the attitude improvement did not extend to non-contacted secondary outgroups and even showed a reverse pattern among young children. This study is among the first developmental research to experimentally demonstrate the relation between interaction with a robot and attitudes toward robots.

CCS Concepts

• **Human-centered computing** → **Laboratory experiments.**

Keywords

human-robot interaction (HRI), child-robot interaction (CRI), intergroup contact theory, attitudes, social robots

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1 Introduction

How do humans feel about machines, and how do those feelings change across development? Scholars have begun to use theories in social psychology to ask such questions, and a few applied *intergroup theories* to examine the potential of robots as tools to improve

human intergroup relationships (e.g., [5, 51]). The current study focuses on human-robot interaction (HRI), conceptualizing HRI as a unique opportunity to investigate how intergroup dynamics observed between different human groups can be extended to interactions between humans and machines. Studies so far have almost exclusively focused on adult populations. Importantly, even young children encounter robots in today's society, and the impact of HRI as well as the effectiveness of intergroup contact are known to vary across development [27, 47, 61]. Further, to fully understand how attitudes toward robots and other outgroups evolve, investigating developmental patterns is inevitable. This study therefore explores whether and how HRI affects intergroup attitudes in two developmental stages — early childhood and middle childhood — using *the intergroup contact theory* as a theoretical framework.

1.1 Intergroup Contact Theory

Intergroup contact theory, originally proposed by Gordon Allport [3] and further refined by Thomas F. Pettigrew [33], posits that positive and meaningful contact with an individual from an outgroup (e.g., different racial, ethnic, or social group) can reduce prejudice toward other members of the outgroup. Subsequent experimental and meta-analytic studies confirmed that a positive, meaningful intergroup interaction with an outgroup member improves attitudes toward the outgroup as a whole (e.g., [32, 35]), primarily through affective intergroup processes such as increased outgroup empathy and decreased intergroup anxiety [36]. The generalization from a member of an outgroup to the whole outgroup is called *primary transfer effect*, and the further generalization of attitude improvement to non-contacted outgroups is *secondary transfer effect* [34]. For instance, after contact with an immigrant, participants in eight European countries showed more positive attitudes toward non-contacted outgroups such as homosexuals [39]. Among limited developmental works, 12-year-old children in Italy, after indirect contact with immigrants, saw non-contacted minority groups (homosexuals and people with disabilities) more positively [54].

1.2 HRI and Intergroup Contact

Robots are commonly perceived as an outgroup, distinct from humans [16], yet human-like stereotypes and characteristics are frequently attributed to robots [28, 29]. Just as humans display negative emotions, attitudes, and reactions toward human outgroups,

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similar responses can emerge toward robots (e.g., [5, 19, 41]). Therefore, techniques used in human intergroup research may also be effective in HRI and offer the potential to improve our relations with robots and other outgroups.

Although the applicability of intergroup contact techniques to HRI has not been thoroughly investigated, existing research touches on the influence of contact with a robot on our attitudes toward robots [59]. A study found both imagined and actual contact with a NAO robot reduced negative emotions toward NAO [60]. Similarly, participants who physically interacted with a robot by listening to its directions and helping it showed marginally greater intention to interact with robots in the future than those who merely observed the robot [57]. In another study, participants engaged in an interactive conversation primarily focusing on exposure or physical contact with the robot [38]. Those who directly interacted with the robot showed less negative explicit and implicit attitudes toward robots. While these studies offer some insights, research rarely extended beyond somewhat superficial contact (e.g., touching, observing), and there has been a need for further exploration of intergroup contact in HRI. In addition, studies that involve talking to the robot focus on a neutral conversation (e.g., [38]) or include only disclosure from the robot, not from the participant (e.g., [17]). Although many of these studies report that even brief exposures to a robot can be effective in improving attitudes, mere exposure with no meaningful contact can adversely strengthen negative attitudes (e.g., [24]).

1.3 Human-Robot Friendship (HRF) Procedure

Research in contact literature suggest that a powerful form of intergroup contact is cross-group friendships [14]. Friendship, by definition, includes positive and reciprocal interactions and self-disclosure. Pettigrew [33] suggests that “friendship potential” is essential in intergroup contact as friendship meets the positive and meaningful contact criteria beyond a superficial interaction. Among the most common is the *fast friendship task*, whereby people from two difference social groups have a conversation involving self-disclosure. Research consistently indicates that this procedure is successful in reducing intergroup anxiety [32], and improving intergroup relations [23].

A pioneering study with young adults [5] applied the fast friendship task to HRI, i.e., the human-robot friendship (HRF) procedure. In the HRF task, the participant and a Pepper robot asked and answered questions in turn (e.g., What would your ideal or perfect life be like?). These participants as well as the control participants (who did not interact with the robot) filled out a survey about their attitudes toward robots and non-contacted human outgroups. The study found the primary transfer effect, showing that HRF was effective in improving attitudes toward robots. As for secondary transfer effects, the authors first conducted a factor analysis on the tested 16 non-contacted outgroups (e.g., elderly, refugees), and identified the Robot, high competence + low warmth (HC-LW), low competence + high warmth (LC-HW), and low competence + low warmth (LC-LW) groups. The consequent mediation analyses found secondary transfer effects for some non-contacted outgroups, in particular the LC-HW groups.

Importantly, findings with adult participants often do not directly translate to younger samples (e.g., [7]). For example, while adult research reflects negative attitudes toward robots [62], a study with 8- to 14-year-olds shows positive attitudes toward robots [48]. Similarly, young children perceive robots more positively and show better attitudes toward robots than adults [8]. A meta-analysis of 20 studies with children aged 3 to 17 also found that young children liked social robots more than older children [43]. Overall, the perception of social robots changes across development, and thus the findings from the adult HRF study [5] must be explored further with younger populations.

2 Present Study

While intergroup contact research consistently demonstrates the effectiveness of the fast friendship procedure, more research is required to understand developmental changes, especially regarding the effects of meeting a robot. The current study thus tests the impact of positive, reciprocal interaction with a social robot in early childhood (ages 4-6) and middle childhood (ages 8-12) and investigates whether the human-robot friendship (HRF) procedure improves children’s attitudes toward robots (primary transfer effect) and non-contacted human outgroups (secondary transfer effect). Our hypotheses are:

- H1. Children in the interaction condition will show improved attitudes toward social robots compared to children in the control (no-interaction) condition, i.e., primary transfer effect.
- H2. Children in the interaction condition will show improved attitudes toward secondary outgroups compared to children in the control (no-interaction) condition, i.e., secondary transfer effect.

2.1 Participants

After parents provide written consent, 68 4- to 6-year-old children ($M = 5.48$ years, $SD = 0.74$; 32 females), and 60 8- to 12-year-old children ($M = 10.10$ years, $SD = 1.05$; 33 females) in Istanbul, Türkiye, participated in the study either at local facilities (e.g., kindergartens) or the Sabancı University campus. The power analysis for planned t -tests resulted in a requirement of at least 52 participants (independent samples t -test with $d = 0.8$, $\alpha = 0.05$, power = 0.80) for each age group.

2.2 Design and Procedure

All children were randomly assigned to the interaction or no-interaction (control) conditions. Children in the interaction condition first participated in the HRF procedure with a NAO robot, and then completed different attitude and behavior measures. Children in the no-interaction condition, which is typically used as a control condition in intergroup research (e.g., [2, 56]), completed the same set of measures without meeting the robot. The robot was teleoperated by a research assistant (RA) in a nearby room using *Choregraphe* (i.e., Wizard-of-Oz technique; [13]). The robot’s speech was generated using *Amazon Polly* (<https://aws.amazon.com/polly>), and adjusted to a higher pitch to mimic child speech.

2.2.1 Human-Robot Friendship (HRF) Procedure. The Research Assistant (RA) individually greeted the children into the room and instructed them to sit facing the NAO robot. The RA asked the NAO and child to introduce themselves to each other and then to ask each other self-disclosure questions (e.g., “What are the things that make you happiest?”). We adapted the four questions from the adult fast friend procedure [4] to match the age of our participants. The number of questions was 8 for younger children and 10 for older children. RA chose questions for young children as they reading the question list is challenging, but older children chose questions on their own from the list. The interaction phase took approximately 10-15 minutes (Figure 1).

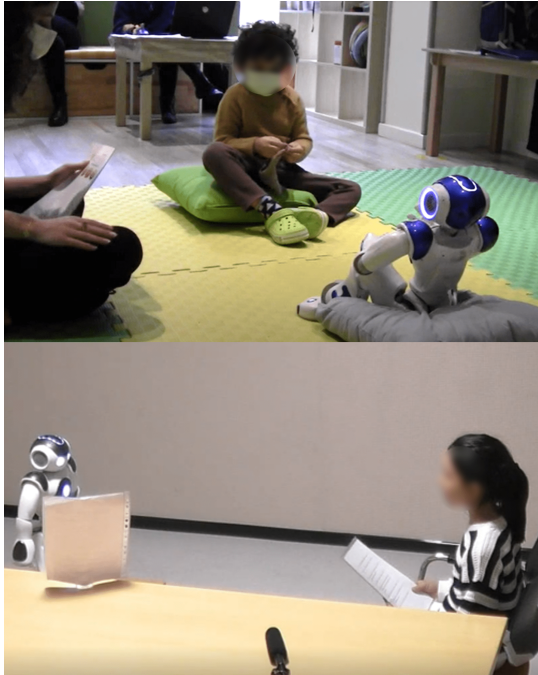


Figure 1: Human-Robot Friendship (HRF) task between the NAO and a child in early (top) and middle childhood (bottom).

2.3 Measures

We tested the same set of constructs – intergroup attitudes, resource allocation, and emotions toward the robot – in both age groups, but slightly different measures were selected for each age group considering their literacy and general cognitive capacity. Older children were additionally tested for their outgroup trust, which was considered to be conceptually too advanced for young children. In addition, as the study was first conducted with young children, minor procedural adjustments (described below) were introduced for older children. All measures were presented in a fixed order, but the order of trials within each measure was randomized across children.

2.3.1 Intergroup Attitudes. The *Multiple-Response Racial Attitude* [1, 6, 15] was used to assess the intergroup attitudes of children. Children were presented with four positive (nice, likable, clean,

good) and four negative (ugly, unpleasant, dirty, bad) traits. Young children were instructed to assign each of these traits either to the picture of the NAO robot, a picture of an ingroup child (gender-matched with the participant), both pictures, or neither of the pictures. Older children completed the rating based on a written description of social robots on a bipolar 5-point scale (1: bad, 5: good), as children in middle childhood can reliably answer on a 5-point scale without visual aids (see [18]).

2.3.2 Resource Allocation. Adapted from [42], we asked children to allocate ten play coins as much as they wished to each of the pictures representing different social groups. The five pictures used for both age groups were: NAO, an ingroup child (White and same gender), a child in a wheelchair, and an elderly person. In addition, younger children were tested with an industrial robot arm, a different-gendered White child, a Black child, an East Asian child, and a chimpanzee, and older children were tested with the Pepper robot. All human outgroups (except the different-gendered White child) were gender-matched with the participant’s gender.

2.3.3 Emotions Toward the Robot. Based on research concerning how emotions relate to attitudes and prejudice [10, 40], we chose nine emotions including excitement, admiration, respect, like, love, jealousy, fear, discomfort, and anger for young children. The child saw a picture of NAO and was asked “How ... would you feel when you see them?” for each emotion. To indicate their answers, younger children used a 3-point face scale consisting of a frown face (“not at all”), a neutral face (“I do not know”), and a smiley face (“very much”). With older children, following the previous study with adults [5], we used on a 7-point scale to test three positive emotions (i.e., sympathy, respect, admiration) toward the robot.

2.3.4 Outgroup Trust. Older children were also tested for their willingness to self-disclose and trust a robot with personal information were also measured (“I would be able to trust a robot as much as any other person,” and “I would be able to trust a robot with personal information about myself”; adapted from [49]), on a scale of 1-7.

3 Results

3.0.1 Intergroup Attitudes. Three independent samples *t*-tests examined young children’s outgroup (NAO), ingroup (same-gendered White child), and intergroup attitudes. Children in the interaction condition ($M = 2.42$, $SD = 2.09$), compared to their peers in the control condition ($M = 1.12$, $SD = 2.78$), showed significantly more positive attitudes toward the outgroup ($t(65) = -2.17$, $p = .034$, $d = -.53$). In contrast, children displayed more negative attitudes toward the ingroup in the interaction condition ($M = -1.36$, $SD = 2.76$), than in the control condition ($M = 0.71$, $SD = 3.03$; $t(65) = 2.92$, $p = .005$, $d = .71$). The intergroup attitude bias was calculated by subtracting the outgroup attitudes from the ingroup attitudes [6], and children in the interaction condition ($M = -3.79$, $SD = 4.41$), compared to the control condition ($M = -0.41$, $SD = 5.03$), had a significant bias favoring the outgroup ($t(65) = 2.92$, $p = .005$, $d = .71$).

Older children also showed positive biases toward robots. Independent samples *t*-tests show that, compared to the control condition, older children in the interaction condition rated robots significantly higher for nice ($t(54) = -2.112$, $p = .039$, $d = -.56$; $M_{interaction}$

= 4.64, $SD = 0.62$; $M_{control} = 4.21$, $SD = 0.88$), likable ($t(54) = -2.25$, $p = .029$, $d = -.60$; $M_{interaction} = 4.61$, $SD = 0.63$; $M_{control} = 4.11$, $SD = .99$), and good ($t(54) = -2.34$, $p = .023$, $d = -.63$; $M_{interaction} = 4.79$, $SD = 0.63$; $M_{control} = 4.29$, $SD = 0.94$), but not for clean ($t(54) = -0.643$, $p = .523$, $d = -.17$; $M_{interaction} = 4.68$, $SD = 0.61$; $M_{control} = 4.57$, $SD = 0.63$).

3.0.2 Resource Allocation. Following the adult HRF study [5], an exploratory factor analysis with Oblimin rotation was conducted to first identify the group structures across the tested social groups [11]. Oblimin rotation was chosen as it best fits the data where some pairs related more closely (e.g., industrial and social robots) than other pairs (e.g., industrial robots and the elderly). The factor analysis revealed three categories (Factor 1: different-gendered White children, children with disabilities, and East Asian children; Factor 2: same-gendered White children (ingroup), industrial robots, and social robots; and Factor 3: Black children and elderly individuals). Independent samples t -tests with the mean allocation scores for each factor indicated that children allocated significantly less resource to the Factor 1 groups in the interaction condition than in the control condition ($t(66) = 3.12$, $p = .003$, $d = .76$; see Figure 2). Similarly, less resource was allocated to the Factor 3 outgroups in the interaction than the control condition ($t(66) = 3.70$, $p < .001$, $d = .90$). No difference between the two conditions was found for Factor 2 ($t(66) = 0.82$, $p = .53$, $d = .15$).

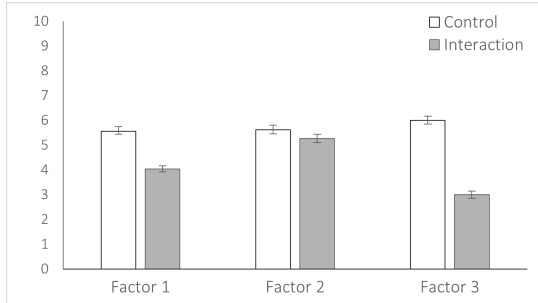


Figure 2: Resources allocated to the outgroups in Factors 1, 2, and 3 in the interaction ($n = 34$) and control conditions ($n = 34$). The error bars indicate standard errors (SE).

For older children, the same factor analysis resulted in Factor 1 (children with disabilities and the elderly) and Factor 2 (NAO and Pepper). Independent samples t -tests with the mean allocation scores indicated that children in the interaction condition did not differ from children in the control condition for either factor (both $ps = .05$; Figure 3). When each outgroup was analyzed separately, however, children in the interaction condition were found to allocate more resources to NAO ($M_{interaction} = 7.10$, $SD = 2.54$; $M_{control} = 5.70$, $SD = 2.74$; $t(58) = -2.05$, $p = .045$, $d = -.53$), but not to any other outgroups (all $ps > .32$).

3.0.3 Emotions Toward Outgroup. The emotions tested with young children were first classified into positive (excitement, admiration, respect, like, love) and negative emotions (jealousy, fear, discomfort, anger). Independent samples t -tests comparing the interaction ($M = 2.47$, $SD = 0.61$) and control conditions ($M = 2.34$, $SD = 0.48$) found

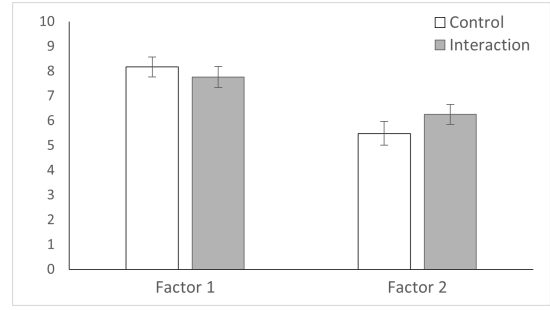


Figure 3: Resources allocated to the outgroups in Factors 1 and 2 in the interaction ($n = 30$) and control conditions ($n = 30$). The error bars indicate standard errors (SE).

no significant difference in positive emotions ($t(66) = -0.99$, $p = .323$, $d = -.24$). Negative emotions also did not differ significantly between the interaction ($M = 1.52$, $SD = 0.57$) and control conditions ($M = 1.41$, $SD = 0.53$; $t(66) = -0.82$, $p = .415$, $d = -.19$).

Older children were asked about three positive emotions only (sympathy, admiration, and respect), and thus each emotion was individually analyzed without grouping them. No significant differences were found between the two conditions regarding sympathy ($t(54) = -1.38$, $p = .175$, $d = -.37$) or admiration ($t(55) = -0.48$, $p = .634$, $d = -.13$). For respect, we observed a trend in which older children in the interaction condition showed higher respect toward social robots than their peers in the control condition, though it did not reach significance ($t(55) = -1.80$, $p = .078$, $d = -.47$).

3.0.4 Outgroup Trust. According to an independent samples t -test, older children showed significantly more trust toward social robots in the interaction condition ($M = 4.81$, $SD = 1.50$) than in the control condition ($M = 3.91$, $SD = 1.84$; $t(56) = -2.03$, $p = .047$, $d = -.53$).

3.1 Discussion

As social robots become increasingly abundant in daily life, it is essential to understand how humans at different developmental stages perceive and interact with them. The present work investigated the impact of human-robot friendship (HRF) procedure, found to be effective with adults [5], on children in early and middle childhood. Overall, our results support H1 (i.e., primary transfer effects), but not H2 (i.e., secondary transfer effects). In other words, as the first developmental investigations on attitudes and behaviors toward robots and other social groups, this study demonstrates that the HRF procedure can improve attitudes toward social robots in both young and older children, but the effects on non-contacted outgroups were limited and showed unexpected patterns in young children.

3.2 HRF in Early Childhood

Young children (4- to 6-year-olds) who interacted with a social robot generally exhibited better attitudes toward social robots than their peers who did not interact with the robot, in line with the adult HRF research [5], as well as other intergroup contact studies on improving children's attitudes toward various social groups (e.g., [9, 55]). Interestingly, however, young children who interacted with

a robot also showed decreased positive attitudes toward the ingroup and favoritism for the outgroup robot, preferring the robot over their peer in a relative status comparison. We speculate that HRF led young children to perceive robots as a particularly admirable outgroup.

Admiration is a favorable reaction toward individuals perceived as superior [50]. The status of an admired outgroup can be elevated with an upward comparison with ingroup [31], and the HRF may have highlighted the positive attributes of robots relative to ingroup. For instance, in a previous study, young children saw robots as smarter than themselves [58]. Although the smartness comparison is about individuals and not groups, a similar thought process can be involved when children compare different social groups [30]. In intergroup research, admiration is known to regulate social hierarchy [44], whereby the status of admired outgroups becomes enhanced relative to ingroup [31].

Since prejudice against robots was low in either condition, robot admiration appears as a plausible explanation for outgroup favoritism in early childhood. Further supporting the interpretation, the factor analysis with resource allocation scores categorized social and industrial robots together with ingroup (Factor 2). Notably, this pattern appears to be unique to young children as robots formed a rather separate group in our middle childhood data and in the previous study with adults [5].

In the resource allocation task, we expected more resources to be allocated to both contacted (i.e., social robots; primary transfer) and non-contacted outgroups (e.g., elderly; i.e., secondary transfer) in the interaction than in the control condition. Conversely, children in the interaction condition allocated fewer resources to secondary outgroups (Factors 1 and 3). We suggest that these children implicitly compared the human outgroups with the robot. This interpretation can also explain why the resource allocation was not different across the conditions for Factor 2, which included social and industrial robots as well as ingroup.

The original HRF study with adults [5] classified the non-contacted outgroups based on competence and warmth and identified secondary transfer effects to groups that are stereotypically low in competence and high in warmth. If children in our study perceived robots as particularly high in warmth and competence, other social groups might have appeared less warm or competent in comparison. Notably, young children do not have clearly separate conceptualizations of competence and warmth [37]. Perhaps children had a vague idea of the Factor 2 groups (ingroup, social robots, industrial robots) being higher in competence or warmth, compared to other tested outgroups (see also [12]).

As for emotions, although past research suggests that positive interaction with a robot can increase positive emotions toward the robot (e.g., [7]), we did not find positive or negative emotions to be different across the conditions. In the current study, young children in both conditions perceived robots positively, with positive emotion ratings above the midpoint of the scale and negative emotion ratings below the midpoint. The high level of positive emotion and the lack of negative emotions also support the aforementioned idea of robots as an admirable outgroup. Nevertheless, we should also be aware of the possibility that the 3-point picture scale might not have sufficient sensitivity, reliability, or validity to detect potential differences between conditions (e.g., [25]).

3.3 HRF in Middle Childhood

As for older children (8- to 12-year-olds), children in the interaction condition rated social robots as higher in the traits of nice, likable, and good, than did their peers in the control condition. Such difference was not observed for cleanliness, assuring that participants considered each individual trait and did not blindly give higher ratings to robots.

Older children's resource allocation resulted in two factors. Factor 1 only included human outgroups (i.e., people with disabilities and the elderly), and Factor 2 included social robots (NAO and Pepper). While the difference between the conditions was not significant for either factor, more resources were allocated to NAO in the interaction than in the control condition, in concert with the expected primary transfer effect.

Previous research found the emergence of the secondary transfer effects in middle childhood, suggesting that positive interaction with an outgroup member can extend to non-contacted outgroups in children [53]. Research also indicates that attitude generalization most commonly occurs when the secondary outgroup is similar to the contacted group [20, 34]. In the current study, attitude generalization was suggested for NAO, but not for other outgroups. The results demonstrate the effectiveness of HRF in children, but only for the outgroup that was similar to the robot they met.

Regarding emotion, past research indicates that positive intergroup contact can increase positive emotions toward the contacted group members (e.g., [46]). The adult HRF study [5] also demonstrated that intimate interaction with a robot can increase positive emotions toward social robots in general. The average respect ratings among older children were higher in the interaction than in the control conditions, in line with these previous studies (see also [7]), but the difference did not reach statistical significance in our data ($p = .08$).

The investigation of trust, in contrast, yielded the expected results. Older children in the interaction condition reported more trust toward social robots than children in the control condition. Past research shows positive contact increases trust toward the contacted group [45], and trust was the most critical finding in the HRF study with adults [5]. Importantly, the contact should be positive and meaningful to be effective, and not every interaction with an outgroup member can increase trust [33]. For instance, Björling et al. [7] reported that observing a robot or talking to a robot did not increase the robot trust of teenagers. The increased trust observed among older children may highlight the importance of self-disclosure from each interaction partner and the advantage of the HRF procedure over other intergroup contact strategies.

3.4 Limitations and Future Directions

This study also offers critical insights into whether and how children should be exposed to social robots. Contrary to H2, secondary transfer effects were not observed in older children and even showed reversed patterns in young children. We speculate that the lack of secondary transfer effects is due to children attributing high status to robots. We did not directly assess children's status evaluation, but children might have seen the status of robots as higher (as can also be seen in younger children categorizing themselves together with the robots), and the status of non-contacted outgroups as lower (or

at least similar to themselves). Indeed, the literature suggests that secondary transfer effects may not emerge when children themselves are of minority status [53]. Apart from the contrast effect, the dis(similarity) might have also played a role. Previous research demonstrated that the more similar the primary and secondary outgroups are, the higher the likelihood of secondary transfer effects [20]. Thus, the lack of secondary transfer effects may be due to robots constituting a group highly dissimilar to the human minority groups we tested, at least in the eyes of children. Research also reports that moderators can block these effects under different circumstances (see [52]). For example, among participants holding high levels of initial prejudice, secondary transfer effects did not occur [21]. Assessing initial levels of prejudice among children and their contact with secondary outgroups may provide further explanations.

The reversed pattern observed in young children also raises a concern about interactions with robots leading to undesirable biases toward human outgroups. As this study is the first of its kind, drawing such a conclusion now would be premature. Yet, the finding certainly proves our earlier point that we cannot just apply what works with adults to children (e.g., [43]). With the increasing use of robots in educational settings, robots are likely to be even more prevalent in children's lives. Intergroup contact in earlier stages of life is known to persist even in adolescence and adulthood [61], suggesting that early contact experiences with robots may shape attitudes toward robots and others later in life. Positive and intimate contact with robots in early childhood may buffer the emergence of negative reactions toward robots, but considering the observed reverse secondary transfer effects, such implementation must be done carefully.

In this regard, we suggest the HRF procedure can be implemented in school settings, given that schools satisfy optimal contact conditions, such as cooperation and support of authorities [3], and allow multiple and sustained structured interactions ("established contact"), which are also critical ingredients of effective contact interventions [33]. Further, though not addressed in this study, research also reports "tertiary transfer effects" [26]—the improvement of socio-cognitive skills such as dispositional empathy and intercultural competence—which may provide additional pedagogical benefits to children. Social robots have been tested at schools (e.g., [22]), but the social and cognitive consequences of such interactions are underexplored. Future research should explore the possible long-term benefits (and drawbacks) of social HRI in diverse settings, and we hope this work encourages more researchers to join the effort.

4 Conclusion

This study is the first to explore the impact of HRF on children's attitudes toward robots and other outgroups. Our findings demonstrate that positive reciprocal interaction with a social robot can improve their attitudes toward robots, but do not generalize to non-contacted outgroups. While the HRF procedure is effective in improving attitudes toward robots even among children, its impact appears to differ across development. The current research also emphasizes the general importance of incorporating a developmental

perspective in discussing how robots are perceived and how the perception can be improved.

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References

- [1] Frances E. Aboud. 2003. The formation of in-group favoritism and out-group prejudice in young children: Are they distinct attitudes? *Developmental Psychology* 39, 1 (Jan. 2003), 48–60. <https://doi.org/10.1037/0012-1649.39.1.48>
- [2] Selen Akay, Berkay Arslan, Sabahat C. Bağcı, and Junko Kanero. 2022. "My Robot Friend": Application of Intergroup Contact Theory in Human-Robot Interaction. In *Proceedings of the 2022 ACM/IEEE International Conference on Human-Robot Interaction (HRI '22)*. IEEE Press, Sapporo, Hokkaido, Japan, 665–668.
- [3] Gordon W. Allport. 1954. *The nature of prejudice*. Addison-Wesley, Oxford, England. Pages: xviii, 537.
- [4] Arthur Aron, Edward Melinat, Elaine N. Aron, Robert Darrin Vallone, and Renee J. Bator. 1997. The experimental generation of interpersonal closeness: A procedure and some preliminary findings. *Personality and Social Psychology Bulletin* 23, 4 (April 1997), 363–377. <https://doi.org/10.1177/0146167297234003>
- [5] Sabahat Cigdem Bağcı, Selan Akay, Ayşe Doğan, and Junko Kanero. 2023. Beyond Interaction: 'Human-Robot (Fast) Friendship' as a Tool to Reduce Prejudice towards Robots and Human Outgroups. <https://doi.org/10.31219/osf.io/tuqgp>
- [6] Michèle D. Birtel, Gian Antonio Di Bernardo, Sofia Stathi, Richard J. Crisp, Alessia Cadamuro, and Loris Vezzali. 2019. Imagining contact reduces prejudice in preschool children. *Social Development* 28, 4 (Nov. 2019), 1054–1073. <https://doi.org/10.1111/sode.12374>
- [7] Elin A. Björklund, Wendy M. Xu, Maria E. Cabrera, and Maya Cakmak. 2019. The Effect of Interaction and Design Participation on Teenagers' Attitudes towards Social Robots. In *2019 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. IEEE, New Delhi, India, 1–7. <https://doi.org/10.1109/RO-MAN46459.2019.8956427>
- [8] Emily R. R. Burdett, Shogo Ikari, and Yo Nakawake. 2022. British Children's and Adults' Perceptions of Robots. *Human Behavior and Emerging Technologies* 2022 (Jan. 2022), 1–16. <https://doi.org/10.1155/2022/3813820>
- [9] L. Cameron, A. Rutland, R. Turner, R. Holman-Nicolas, and C. Powell. 2011. Changing attitudes with a little imagination: Imagined contact effects on young children's intergroup bias. *Anales de Psicología* 27, 3 (2011), 708–717.
- [10] Catherine A. Cottrell, David A.R. Richards, and Austin Lee Nichols. 2010. Predicting policy attitudes from general prejudice versus specific intergroup emotions. *Journal of Experimental Social Psychology* 46, 2 (March 2010), 247–254. <https://doi.org/10.1016/j.jesp.2009.10.008>
- [11] Charles Crawford. 1975. A Comparison of The Direct Oblimin And Primary Parsimony Methods of Oblique Rotation. *Brit. J. Math. Statist. Psych.* 28, 2 (Nov. 1975), 201–213. <https://doi.org/10.1111/j.2044-8317.1975.tb00563.x>
- [12] Amy J.C. Cuddy, Susan T. Fiske, and Peter Glick. 2008. Warmth and Competence as Universal Dimensions of Social Perception: The Stereotype Content Model and the BIAS Map. In *Advances in Experimental Social Psychology*. Vol. 40. Elsevier, 61–149. [https://doi.org/10.1016/S0065-2601\(07\)00002-0](https://doi.org/10.1016/S0065-2601(07)00002-0)
- [13] N. Dahlbäck, A. Jönsson, and L. Ahrenberg. 1993. Wizard of Oz studies — why and how. *Knowledge-Based Systems* 6, 4 (Dec. 1993), 258–266. [https://doi.org/10.1016/0950-7051\(93\)90017-N](https://doi.org/10.1016/0950-7051(93)90017-N)
- [14] Kristin Davies, Linda R. Tropp, Arthur Aron, Thomas F. Pettigrew, and Stephen C. Wright. 2011. Cross-group friendships and intergroup attitudes: A meta-analytic review. *Personality and Social Psychology Review* 15, 4 (Nov. 2011), 332–351. <https://doi.org/10.1177/1088868311411103>
- [15] Anna Beth Doyle and Frances E. Aboud. 1995. A Longitudinal Study of White Children's Racial Prejudice as a Social-Cognitive Development. *Merrill-Palmer Quarterly* 41, 2 (1995), 209–228. <http://www.jstor.org/stable/23090532> Publisher: Wayne State University Press.
- [16] Friederike Eyssel and Frank Hegel. 2012. (S)he's got the look: Gender stereotyping of robots. *Journal of Applied Social Psychology* 42, 9 (2012), 2213–2230. <https://doi.org/10.1111/j.1559-1816.2012.00937.x> <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1559-1816.2012.00937.x>

- [17] F. Eyssell, R. Wullenkord, and V. Nitsch. 2017. The role of self-disclosure in human-robot interaction. In *2017 26th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*. IEEE, Lisbon, 922–927. <https://doi.org/10.1109/ROMAN.2017.8172413>
- [18] G B Fabiola, L Iwin, L M Jennifer, and V V Zaira. 2012. The effect of the number of answer choices on the psychometric properties of stress measurement in an instrument applied to children. *Evaluate* 12 (2012), 43–59.
- [19] Brad A. Haggadone, Jaime Banks, and Kevin Koban. 2021. Of robots and robotkind: Extending intergroup contact theory to social machines. *Communication Research Reports* 38, 3 (May 2021), 161–171. <https://doi.org/10.1080/08824096.2021.1909551>
- [20] Jake Harwood, Stefania Paolini, Nick Joyce, Mark Rubin, and Analisa Arroyo. 2011. Secondary transfer effects from imagined contact: Group similarity affects the generalization gradient. *British Journal of Social Psychology* 50, 1 (March 2011), 180–189. <https://doi.org/10.1348/014466610X524263>
- [21] Inga Jasinskaja-Lahti, Loris Vezzali, Mona Ranta, Maria Giuseppina Pacilli, Mauro Giacomantonio, and Stefano Pagliaro. 2021. Conditional secondary transfer effect: The moderating role of moral credentials and prejudice. *Group Processes & Intergroup Relations* 24, 7 (Oct. 2021), 1219–1237. <https://doi.org/10.1177/1368430220940401> Publisher: SAGE Publications Ltd.
- [22] T. Kanda, R. Sato, N. Saiwaki, and H. Ishiguro. 2007. A Two-Month Field Trial in an Elementary School for Long-Term Human-Robot Interaction. *IEEE Transactions on Robotics* 23, 5 (Oct. 2007), 962–971. <https://doi.org/10.1109/TRO.2007.904904>
- [23] Patrick F. Kotzur and Ulrich Wagner. 2021. The dynamic relationship between contact opportunities, positive and negative intergroup contact, and prejudice: A longitudinal investigation. *Journal of Personality and Social Psychology* 120, 2 (Feb. 2021), 418–442. <https://doi.org/10.1037/pspi0000258>
- [24] James Laurence, Katharina Schmid, and Miles Hewstone. 2018. Ethnic Diversity, Inter-group Attitudes and Countervailing Pathways of Positive and Negative Inter-group Contact: An Analysis Across Workplaces and Neighbourhoods. *Social Indicators Research* 136, 2 (April 2018), 719–749. <https://doi.org/10.1007/s11205-017-1570-z>
- [25] Luis M. Lozano, Eduardo García-Cueto, and José Muñiz. 2008. Effect of the Number of Response Categories on the Reliability and Validity of Rating Scales. *Methodology* 4, 2 (Jan. 2008), 73–79. <https://doi.org/10.1027/1614-2241.4.2.73>
- [26] Rose Meleady, Richard J. Crisp, Gordon Hodson, and Megan Earle. 2019. On the generalization of intergroup contact: A taxonomy of transfer effects. *Current Directions in Psychological Science* 28, 5 (Oct. 2019), 430–435. <https://doi.org/10.1177/0963721419848682>
- [27] Christine E. Merrilees, Laura K. Taylor, Madeline Klotz, Marcie C. Goeke-Morey, Peter Shirlow, and E. Mark Cummings. 2023. Timing is everything: Developmental changes in the associations between intergroup contact and bias. *International Journal of Behavioral Development* 47, 3 (May 2023), 243–252. <https://doi.org/10.1177/01650254221146409>
- [28] Clifford Nass and Youngme Moon. 2000. Machines and Mindlessness: Social Responses to Computers. *Journal of Social Issues* 56, 1 (Jan. 2000), 81–103. <https://doi.org/10.1111/0022-4537.00153>
- [29] Clifford Nass, Youngme Moon, and Paul Carney. 1999. Are People Polite to Computers? Responses to Computer-Based Interviewing Systems¹. *Journal of Applied Social Psychology* 29, 5 (May 1999), 1093–1109. <https://doi.org/10.1111/j.1559-1816.1999.tb00142.x>
- [30] Diana Onu, Thomas Kessler, Daniela Andonovska-Trajkovska, Immo Fritsche, Georgia R. Midson, and Joanne R. Smith. 2016. Inspired by the outgroup: A social identity analysis of intergroup admiration. *Group Processes & Intergroup Relations* 19, 6 (Nov. 2016), 713–731. <https://doi.org/10.1177/1368430216629811>
- [31] Diana Onu, Joanne R. Smith, and Thomas Kessler. 2015. Intergroup emulation: An improvement strategy for lower status groups. *Group Processes & Intergroup Relations* 18, 2 (March 2015), 210–224. <https://doi.org/10.1177/1368430214556698>
- [32] Elizabeth Page-Gould, Rodolfo Mendoza-Denton, and Linda R. Tropp. 2008. With a little help from my cross-group friend: Reducing anxiety in intergroup contexts through cross-group friendship. *Journal of Personality and Social Psychology* 95, 5 (2008), 1080–1094. <https://doi.org/10.1037/0022-3514.95.5.1080>
- [33] Thomas F. Pettigrew. 1998. Intergroup contact theory. *Annual Review of Psychology* 49, 1 (Feb. 1998), 65–85. <https://doi.org/10.1146/annurev.psych.49.1.65>
- [34] Thomas F. Pettigrew. 2009. Secondary Transfer Effect of Contact: Do Intergroup Contact Effects Spread to Noncontacted Outgroups? *Social Psychology* 40, 2 (Jan. 2009), 55–65. <https://doi.org/10.1027/1864-9335.40.2.55>
- [35] Thomas F. Pettigrew and Linda R. Tropp. 2006. A meta-analytic test of intergroup contact theory. *Journal of Personality and Social Psychology* 90, 5 (2006), 751–783. <https://doi.org/10.1037/0022-3514.90.5.751>
- [36] Thomas F. Pettigrew and Linda R. Tropp. 2008. How does intergroup contact reduce prejudice? Meta-analytic tests of three mediators. *European Journal of Social Psychology* 38, 6 (2008), 922–934. <https://doi.org/10.1002/ejsp.504> eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1002/ejsp.504>
- [37] Gina Roussos and Yarrow Dunham. 2016. The development of stereotype content: The use of warmth and competence in assessing social groups. *Journal of Experimental Child Psychology* 141 (Jan. 2016), 133–144. <https://doi.org/10.1016/j.jecp.2015.08.009>
- [38] Marina Sarda Gou, Thomas L. Webb, and Tony Prescott. 2021. The effect of direct and extended contact on attitudes towards social robots. *Heliyon* 7, 3 (March 2021), e06418. <https://doi.org/10.1016/j.heliyon.2021.e06418>
- [39] Katharina Schmid, Miles Hewstone, Beate Küpper, Andreas Zick, and Ulrich Wagner. 2012. Secondary Transfer Effects of Intergroup Contact: A Cross-National Comparison in Europe. *Social Psychology Quarterly* 75, 1 (March 2012), 28–51. <https://doi.org/10.1177/0190272511430235>
- [40] Charles R. Seger, Ishani Banerji, Sang Hee Park, Eliot R. Smith, and Diane M. Mackie. 2017. Specific emotions as mediators of the effect of intergroup contact on prejudice: findings across multiple participant and target groups. *Cognition and Emotion* 31, 5 (July 2017), 923–936. <https://doi.org/10.1080/02699931.2016.1182893>
- [41] Eliot R. Smith, Steven Sherrin, Marlana R. Fraune, and Selma Šabanović. 2020. Positive Emotions, More Than Anxiety or Other Negative Emotions, Predict Willingness to Interact With Robots. *Personality and Social Psychology Bulletin* 46, 8 (Aug. 2020), 1270–1283. <https://doi.org/10.1177/0146167219900439>
- [42] Darren A. Spielman. 2000. Young Children, Minimal Groups, and Dichotomous Categorization. *Personality and Social Psychology Bulletin* 26, 11 (Nov. 2000), 1433–1441. <https://doi.org/10.1177/0146167200263010>
- [43] Rebecca Stower, Natalia Calvo-Barajas, Ginevra Castellano, and Arvid Kappas. 2021. A Meta-analysis on Children's Trust in Social Robots. *International Journal of Social Robotics* 13, 8 (Dec. 2021), 1979–2001. <https://doi.org/10.1007/s12369-020-00736-8>
- [44] Joseph Sweetman, Russell Spears, Andrew G. Livingstone, and Antony S.R. Manstead. 2013. Admiration regulates social hierarchy: Antecedents, dispositions, and effects on intergroup behavior. *Journal of Experimental Social Psychology* 49, 3 (May 2013), 534–542. <https://doi.org/10.1016/j.jesp.2012.10.007>
- [45] Tania Tam, Miles Hewstone, Jared Kenworthy, and Ed Cairns. 2009. Intergroup Trust in Northern Ireland. *Personality and Social Psychology Bulletin* 35, 1 (Jan. 2009), 45–59. <https://doi.org/10.1177/0146167208325004>
- [46] Lingjie Tang, Chang'an Zhang, and Zhifang Liu. 2024. How can previous intergroup contact predict willingness for intergroup interaction? The mediating role of specific intergroup emotions. *Asian Journal of Social Psychology* (March 2024), ajsp.12607. <https://doi.org/10.1111/ajsp.12607>
- [47] Linda R. Tropp, Fiona White, Christina L. Rucinski, and Colin Tredoux. 2022. Intergroup Contact and Prejudice Reduction: Prospects and Challenges in Changing Youth Attitudes. *Review of General Psychology* 26, 3 (Sept. 2022), 342–360. <https://doi.org/10.1177/10892680211046517>
- [48] Fang-Wu Tung. 2016. Child Perception of Humanoid Robot Appearance and Behavior. *International Journal of Human-Computer Interaction* 32, 6 (June 2016), 493–502. <https://doi.org/10.1080/10447318.2016.1172808>
- [49] Rhiannon N. Turner, Keon West, and Zara Christie. 2013. Out-group trust, intergroup anxiety, and out-group attitude as mediators of the effect of imagined intergroup contact on intergroup behavioral tendencies: Mediators and consequences of imagined contact. *Journal of Applied Social Psychology* 43 (June 2013), E196–E205. <https://doi.org/10.1111/jasp.12019>
- [50] Niels Van De Ven, Marcel Zeelenberg, and Rik Pieters. 2011. Why Envy Outperforms Admiration. *Personality and Social Psychology Bulletin* 37, 6 (June 2011), 784–795. <https://doi.org/10.1177/0146167211400421>
- [51] Eric J. Vanman and Arvid Kappas. 2019. "Danger, Will Robinson!" The challenges of social robots for intergroup relations. *Social and Personality Psychology Compass* 13, 8 (Aug. 2019). <https://doi.org/10.1111/spc3.12489>
- [52] Loris Vezzali, Gian Antonio Di Bernardo, Veronica Margherita Cocco, Sofia Stathi, and Dora Capozza. 2021. Reducing prejudice in the society at large: A review of the secondary transfer effect and directions for future research. *Social and Personality Psychology Compass* 15, 3 (March 2021). <https://doi.org/10.1111/spc3.12583>
- [53] Loris Vezzali, Gian Antonio Di Bernardo, Sofia Stathi, Alessia Cadamuro, Barbara Lásticová, and Simona Andračiková. 2018. Secondary transfer effect among children: The role of social dominance orientation and outgroup attitudes. *British Journal of Social Psychology* 57, 3 (July 2018), 547–566. <https://doi.org/10.1111/bjso.12248>
- [54] Loris Vezzali and Dino Giovannini. 2012. Secondary Transfer Effect of Intergroup Contact: The Role of Intergroup Attitudes, Intergroup Anxiety and Perspective Taking. *Journal of Community & Applied Social Psychology* 22, 2 (March 2012), 125–144. <https://doi.org/10.1002/casp.1103>
- [55] Loris Vezzali, Miles Hewstone, Dora Capozza, Elena Trifiletti, and Gian Antonio Di Bernardo. 2017. Improving Intergroup Relations with Extended Contact among Young Children: Mediation by Intergroup Empathy and Moderation by Direct Intergroup Contact. *Journal of Community & Applied Social Psychology* 27, 1 (Jan. 2017), 35–49. <https://doi.org/10.1002/casp.2292>
- [56] Loris Vezzali, Sofia Stathi, Richard J. Crisp, and Dora Capozza. 2015. Comparing direct and imagined intergroup contact among children: Effects on outgroup stereotypes and helping intentions. *International Journal of Intercultural Relations* 49 (Nov. 2015), 46–53. <https://doi.org/10.1016/j.ijintrel.2015.06.009>
- [57] Kathryn Wallisch, Marlana Fraune, Selma Šabanović, Steven Sherrin, and Eliot Smith. 2018. Getting to Know You: Relationship Between Intergroup Contact and Willingness to Interact. In *Companion of the 2018 ACM/IEEE International*

- Conference on Human-Robot Interaction*. ACM, Chicago IL USA, 275–276. <https://doi.org/10.1145/3173386.3177017>
- [58] Randi Williams, Hae Won Park, and Cynthia Breazeal. 2019. A is for Artificial Intelligence: The Impact of Artificial Intelligence Activities on Young Children’s Perceptions of Robots. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. ACM, Glasgow Scotland Uk, 1–11. <https://doi.org/10.1145/3290605.3300677>
- [59] Ricarda Wullenkord and Friederike Eyssel. 2019. Imagine how to behave: the influence of imagined contact on human–robot interaction. *Philosophical Transactions of the Royal Society B: Biological Sciences* 374, 1771 (April 2019), 20180038. <https://doi.org/10.1098/rstb.2018.0038>
- [60] Ricarda Wullenkord, Marlena R. Fraune, Friederike Eyssel, and Selma Sabanovic. 2016. Getting in Touch: How imagined, actual, and physical contact affect evaluations of robots. In *2016 25th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*. IEEE, New York, NY, USA, 980–985. <https://doi.org/10.1109/ROMAN.2016.7745228>
- [61] Ralf Wölfer, Katharina Schmid, Miles Hewstone, and Maarten Van Zalk. 2016. Developmental Dynamics of Intergroup Contact and Intergroup Attitudes: Long-Term Effects in Adolescence and Early Adulthood. *Child Development* 87, 5 (Sept. 2016), 1466–1478. <https://doi.org/10.1111/cdev.12598>
- [62] Chung-En Yu. 2020. Humanlike robots as employees in the hotel industry: Thematic content analysis of online reviews. *Journal of Hospitality Marketing & Management* 29, 1 (Jan. 2020), 22–38. <https://doi.org/10.1080/19368623.2019.1592733>